$$\hat{\theta} = \underset{\theta}{\operatorname{argmax}}\ p(\theta)\, p(x|\theta) = \ln\left[\underset{\theta}{\operatorname{argmax}}\ p(\theta)p(x|\theta)\right] = \underset{\theta}{\operatorname{argmax}}\ \ln[p(\theta)] + \ln[p(x|\theta)]$$

$$p(\theta) = \frac{\sqrt{\nu}}{\sigma\sqrt{2\pi}} \frac{\beta^{\alpha}}{\Gamma(\alpha)} \left(\frac{1}{\sigma^2}\right)^{\alpha+1} \exp\left[-\frac{2\beta + \nu(\mu_0-\mu)^2}{2\sigma^2}\right]$$

$$\ln p(\theta) = \ln\left[\frac{\sqrt{\nu}}{\sigma\sqrt{2\pi}} \frac{\beta^{\alpha}}{\Gamma(\alpha)} \left(\frac{1}{\sigma^2}\right)^{\alpha+1}\right] - \frac{2\beta + \nu(\mu_0-\mu)^2}{2\sigma^2} =$$

$$= \ln[\sqrt{\nu}] - \ln[\sigma\sqrt{2\pi}] + \ln\left[\frac{\beta^{\alpha}}{\Gamma(\alpha)}\right] + \ln[1^{\alpha+1}] - \ln[\sigma^{2\alpha+2}] - \frac{2\beta + \nu(\mu_0-\mu)^2}{2\sigma^2}$$

$$p(x|\theta) = \prod_{i=1}^{N}\left[\frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\frac{(x_i-\mu)^2}{\sigma^2}\right]\right]$$

$$\ln p(x|\theta) = \sum_{i=1}^{N}\left[-\ln[\sigma\sqrt{2\pi}]\right] - \sum_{i=1}^{N}\frac{(x_i-\mu)^2}{2\sigma^2}$$

<u>odvození σ^2</u>

$$\frac{\partial \ln p(\theta)}{\partial\sigma} = -\frac{1}{\sigma} - \frac{2\alpha-2}{\sigma} + 2\cdot\frac{2\beta+\nu(\mu_0-\mu)^2}{2\sigma^3} = -\frac{2\alpha-3}{\sigma} + \frac{2\beta+\nu(\mu_0-\mu)^2}{\sigma^3}$$

$$\frac{\partial \ln p(x|\theta)}{\partial\sigma} = -\frac{N}{\sigma} - \sum_{i=1}^{N} -2\cdot\frac{(x_i-\mu)^2}{2\sigma^3} = -\frac{N}{\sigma} + \sum_{i=1}^{N}\frac{(x_i-\mu)^2}{\sigma^3}$$

$$\frac{\partial \ln p(\theta)}{\partial\sigma} + \frac{\partial \ln(p(x|\theta))}{\partial\sigma} = 0$$

$$-\frac{N+1}{\hat{\sigma}} - \frac{2\alpha+2}{\hat{\sigma}} + \frac{2\beta+\nu(\mu_0-\mu)^2}{\hat{\sigma}^3} + \frac{\sum_{i=1}^{N}(x_i-\mu)^2}{\hat{\sigma}^3} = 0$$

$$\hat{\sigma}^2(N+1+2\alpha+2) = 2\beta + \nu(\mu_0-\mu)^2 + \sum_{i=1}^{N}(x_i-\mu)^2$$

$$\hat{\sigma}^2 = \frac{2\beta + \nu(\mu_0-\mu)^2 + \sum_{i=1}^{N}(x_i-\mu)^2}{N+3+2\alpha}$$

<u>odvození $\hat{\mu}$</u>

$$\frac{\partial \ln p(\theta)}{\partial\mu} = \frac{-2\nu(\mu_0-\mu)}{2\sigma^2} = -\frac{\nu(\mu_0-\mu)}{\sigma^2}$$

$$\frac{\partial \ln p(x|\theta)}{\partial\mu} = \sum_{i=1}^{N}\frac{-2(x_i-\mu)}{2\sigma^2} = \sum_{i=1}^{N} -\frac{x_i-\mu}{\sigma^2}$$

$$\frac{\partial \ln p(\theta)}{\partial \mu} + \frac{\partial \ln p(x|\theta)}{\partial \mu} = 0$$

$$-\frac{\nu(\mu_0 - \hat{\mu})}{\sigma^2} - \sum_{i=1}^{N} \frac{x_i - \hat{\mu}}{\sigma^2} = 0$$

$$\hat{\mu}(\nu + N) = \nu \mu_0 + \sum_{i=1}^{N} x_i$$

$$\hat{\mu} = \frac{\nu \mu_0 + \sum_{i=1}^{N} x_i}{\nu + N}$$